

AE4L GATE Self Study Guide

Kakade, D.

STAR Rocketry
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Chapter 1

Syllabus For Gate AE 2027

1.1 Maths

1.1.1 Core Topics

Linear Algebra: Vector algebra, matrix algebra, systems of linear equations, rank of a matrix; Eigenvalues and eigenvectors.

Calculus: Functions of single variable, limits, continuity and differentiability, chain rule, maxima and minima, Integration; Functions of several variables, partial derivatives, gradient, divergence and curl, directional derivatives; Line, surface and volume integrals. Theorems of Stokes, Gauss and Green.

Differential Equations: First order linear ordinary differential equations; Higher order linear ODEs with constant coefficients; Classification of partial differential equations; *Solution to:* wave equation, Laplace equation, heat equation using separation of variables methods.

Numerical Methods: Numerical solutions for linear and nonlinear algebraic equations by bisection, Newton-Raphson method; basic numerical differentiation; numerical integration by trapezoidal and Simpson's rules; linear regression and least squares method; linear interpolation.

1.1.2 Special Topics

Fourier series; complex numbers, analytic functions and Cauchy-Riemann equations. Basics of probability and statistics: Bayes theorem, mean, median and mode; variance; binomial, normal and Poisson distribution.

1.2 Aerodynamics

1.2.1 Core Topics

Basic fluid mechanics: Fluid kinematics, streamline, streakline, pathline; conservation laws: mass, linear momentum and energy (integral and differential form); dimensional analysis and dynamic similarity; incompressibility conditions; Newtonian fluids.

Incompressible Flow

Elementary ideas of viscous flows: Hagen-Poiseuille flow, Couette flow, basic concepts in boundary layers, boundary layer thickness. Two-dimensional potential flow theory: sources, sinks, doublets, point vortex and their superposition, Bernoulli's equation.

Airfoils: Airfoil nomenclature; aerodynamic coefficients: lift, drag and moment; Kutta-Joukowski theorem; thin airfoil theory, Kutta condition, starting vortex;

Wings: Finite wing theory: induced drag, Prandtl lifting line theory; critical and drag divergence Mach numbers.

Compressible flows

Basic concepts of compressibility, one-dimensional compressible flows, isentropic flows, normal and oblique shocks, Prandtl-Meyer flow; flow through nozzles and diffusers.

1.2.2 Special Topics

Fanno flow; Rayleigh flow; pressure measurements using U-tube manometers and Pitot probe.

1.3 Gas Turbines

1.3.1 Core Topics

Basics of thermodynamics.

Aerothermodynamics of aircraft engines: Thrust, efficiency, range. Brayton cycle.

Engine performance: Ramjet, turbojet, turbofan, turboprop and turboshaft engines; afterburners.

Aerothermodynamics of non-rotating propulsion components: Intakes, nozzles.

Gas turbine combustor: Types and configurations; combustion, stoichiometric fuel-to-air ratio.

Turbomachinery

Axial compressors: Angular momentum, work and compression, characteristic performance of a single axial compressor stage, efficiency of the compressor and degree of reaction, multi-staging; *Centrifugal compressor:* Stage dynamics, inducer, impeller and diffuser; Axial turbines: stage performance.

1.3.2 Special Topics

Turbine blade cooling, compressor-turbine matching.

1.4 Rocket Propulsion

Rockets thrust equation and specific impulse, rocket performance; multi-staging; chemical rockets; performance of solid and liquid propellant rockets.

1.5 Flight Mechanics

Atmosphere Properties of standard atmosphere: Classification of aircraft. Airplane (fixed wing aircraft) configuration and various parts; Pressure altitude; equivalent, calibrated, indicated air speeds;

Primary flight instruments: Altimeter, air speed indicator, vertical speed indicator, turn-bank indicator.

Basics: Aerodynamic forces and moments; angle of attack; sideslip; High-lift devices; roll, pitch and yaw controls.

Aircraft Performance

C_L vs. α curve, drag polar; take-off and landing; steady climb and descent; absolute and service ceiling; range and endurance, load factor, turning flight, V-n diagram. Winds: head, tail and cross winds.

Static stability

Stability and control derivatives; longitudinal stick fixed and free stability; horizontal tail position and size; directional stability, vertical tail position and size; lateral stability; wing dihedral, sweep and position; hinge moments, stick forces. Linear momentum and angular momentum balance for rigid bodies.

1.5.1 Special Topics

Equations of motion; Euler angles; decoupling of longitudinal and lateral-directional dynamics; longitudinal modes; lateral-directional modes.

1.6 Orbital Mechanics

1.6.1 Core Topics

Central force motion, Keplerian orbits, Kepler's laws; escape velocity.

1.6.2 Special Topics

Hohmann orbital transfers.

1.7 Aircraft Structures

1.7.1 Core Topics

Strength Of Materials: Stress and strain, stress-strain curves of steel and Aluminum; stresses and deflections in statically determinate and indeterminate linear elastic trusses, bars, beams, and

shafts; two-dimensional transformations of stresses and strains, Mohr's circle, principal stresses and strains; combined loading, failure criteria: maximum stress, Tresca, von Mises; strain energy; Castigliano's principles; three-dimensional Hooke's law; plane stress and strain; Euler buckling of columns.

Flight Vehicle Structure: Torsion, bending and shear of open and closed thin-walled sections: symmetric and unsymmetrical cross-sections; loads on aircraft.

Mechanical Vibration: Free and forced vibrations of undamped and damped SDOF systems; free vibrations of undamped 2-DOF systems.

1.7.2 Special Topics

Aero-Elasticity: Equilibrium and compatibility equations in two-dimensional elasticity. [Airy's stress function (removed)]

1.8 Aptitude

Verbal: Basic English grammar: tenses, articles, adjectives, prepositions, conjunctions, verb-noun agreement, and other parts of speech. Basic vocabulary: words, idioms, and phrases in context. Reading and comprehension, Narrative sequencing.

Quant: Data interpretation: data graphs (bar graphs, pie charts, and other graphs representing data), 2-and 3-dimensional plots, maps, and tables. Numerical computation and estimation: ratios, percentages, powers, exponents and logarithms, permutations and combinations, and series. Mensuration and geometry, Elementary statistics and probability.

Analytical: Logic: deduction and induction, Analogy, Numerical relations and reasoning.

Spatial: Transformation of shapes: translation, rotation, scaling, mirroring, assembling, and grouping paper folding, cutting, and patterns in 2 and 3 dimensions.

Chapter 2

Reference Books

2.1 Engineering Mathematics

1. Advanced Engineering Mathematics by Erwin Kreyszig
2. Advanced Engineering Mathematics by H. K. Dass
3. Higher Engineering Mathematics by B. S. Grewal

2.2 Fluid Mechanics

1. Fluid Mechanics: Fundamentals and Applications by Yunus A. Cengel, John M. Cimbala
2. Fluid Mechanics by Frank M. White
3. Fluid Mechanics by RK Bansal (for basics and more solved problems)

2.3 Aero / Gas Dynamics

1. Fundamentals of Gas Dynamics by Robert D. Zucker, Oscar Biblarz
2. Fundamentals of Aerodynamics by John Anderson
3. Modern Compressible Flow: With Historical Perspective by John Anderson
4. Fundamentals Of Compressible Flow by SM Yahya
5. Aerodynamics for Engineering students by E. L. Houghton, N. B. Carruthers
6. Gas Dynamics by Ethirajan Rathakrishnan

2.4 Aircraft Propulsion

1. Gas Turbine Theory by Cohen, H., Rogers
2. Gas Turbines by V. Ganesan
3. Aircraft Propulsion and Gas Turbine Engines by Ahmed F. El-Sayed
4. Aircraft Propulsion by Saeed Farokhi

2.5 Rocket Propulsion

1. Rocket Propulsion Elements by George P. Sutton
2. Propellants and Explosives by Naminosuke Kubota
3. Rocket Propulsion by K Ramamurthi

2.6 Flight Mechanics

1. Aircraft Performance and Design by John Anderson
2. Flight Stability and Automatic Control by R. C. Nelson
3. Airplane Performance, Stability and Control by Courtland D. Perkins, Robert E. Hage
4. Introduction to Flight by John Anderson
5. Aerodynamics by Clancy

2.7 Space Mechanics

1. Orbital Mechanics for Engineering Students by Howard D. Curtis
2. Fundamentals of Astrodynamics by Donald D. Mueller, Jerry White, Roger R. Bate
3. Spaceflight Dynamics by William E. Wiesel
4. Orbital Mechanics by John E. Prussing, Bruce A. Conway
5. Analytical Mechanics of Space Systems by Hanspeter Schaub, John L. Junkins

2.8 Strength of Materials

1. Strength of Materials by R.K. Rajput
2. Mechanics of Materials by James M. Gere
3. Strength Of Materials Parts I and II by Timoshenko S.
4. Strength of Materials by S. Ramamrutham

2.9 Vibrations

1. Mechanical Vibrations by S.S Rao
2. Dynamics of Structures by Jagmohan L. Humar
3. Mechanical Vibrations by V.P Singh

2.10 Structures

1. Aircraft Structures for Engineering Students By T.H.G. Megson
2. Analysis and Design of Flight Vehicle Structures by E. F Bruhn
3. Mechanics of Aircraft Structures by C. T. Sun
4. Analysis of Aircraft Structures: An Introduction by Bruce k. Donaldson

2.11 Heat And Mass Transfer

1. Heat and Mass Transfer by R.K. Rajput
2. Heat and Mass Transfer by JP holman

2.12 Previous Year Questions

GK Publisher

2.13 Other Ways To Learn

1. MIT OCW / NPTEL - Assignments and Problem Sets.
2. Aerospace Engineering - GATE exam Subject review notes with simplified conceptual explanation and problem solving tips - An anonymous IIT Graduate.

Chapter 3

Unit Conversions and Constants

3.1 Unit Conversions

$$1atm = 1.01bar = 1.01 \times 10^5 N/m^2$$

$$1bar = 10^5 Pa = 10^5 N/m^2$$

$$1feet = 0.3048m$$

$$1slug = 14.59kg$$

$$1lb = 0.453kg$$

$$1mile = 1.609km$$

$$1K = 1.8^\circ R$$

$$59^\circ R = 288K$$

$$0^\circ F = 459.67^\circ R$$

$$1kgf = 9.81N$$

$$1N = 0.102kgf$$

$$1hp = 746W$$

3.2 Constants

$$\gamma_{air} = 1.4$$

$$\gamma_{gas} = 1.33$$

$$M_{air} = 28.9kgmol^{-1}$$

$$C_{p, air} = 1004Jkg^{-1}K^{-1}$$

$$C_{p, gas} = 1148Jkg^{-1}K^{-1}$$

$$\bar{R} = 8314Jkg^{-1}mol^{-1}$$

$$R = 287 J kg^{-1}K^{-1}$$

$$P_{sea level} = 1atm = 1.01 \times 10^5 Nm^{-2}$$

$$\rho_{sea level} = 1.23 kg m^{-3}$$

$$T_{sea level} = 15 deg C = 288K$$

Chapter 4

Maths

4.1 Nomenclature

Symbol	Meaning	Unit
ρ	density	kgm^3

Chapter 5

Aerodynamics

5.1 Nomenclature

Symbol	Meaning	Unit
ρ	density	kgm^3

5.2 Fluid Mechanics

This section follows Ref. [1] as the main reference.

5.3 Potential Flow Theory and Viscous Flow

This section onwards Ref. [2] will be followed as the main reference.

5.4 Flow Over Airfoils

5.5 Finite Wing Theory

5.6 Compressible Flow

5.6.1 Normal Shock

5.6.2 Oblique Shock

5.6.3 Expansion Fan

5.7 Nozzles and Diffusers

5.8 Fanno Flow, Rayleigh Flow

5.9 Boundary Layers

Chapter 6

Gas Turbine

6.1 Nomenclature

Symbol	Meaning	Unit
ρ	density	kgm^3

Chapter 7

Rocket Propulsion

7.1 Nomenclature

Symbol	Meaning	Unit
ρ	density	kgm^3

Chapter 8

Flight Mechanics

8.1 Nomenclature

Symbol	Meaning	Unit
ρ	density	kgm^3

Chapter 9

Orbital Mechanics

9.1 Nomenclature

Symbol	Meaning	Unit
ρ	density	kgm^3

Chapter 10

Aircraft Structures

10.1 Nomenclature

Symbol	Meaning	Unit
ρ	density	kgm^3

References

- [1] Yunus A. Çengel and John M. Cimbala. *Fluid mechanics: fundamentals and applications*. McGraw-Hill, Singapore, fourth edition in si units edition, 2020. ISBN 978-981-315-788-0.
- [2] John David Anderson and Christopher P. Cadou. *Fundamentals of aerodynamics*. McGraw Hill series in aeronautical and aerospace engineering. McGraw Hill, New York, seventh edition, international student edition edition, 2024. ISBN 978-1-266-07644-2.